Guthula Guna Vardhan

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Objective

Motivated Computer Science graduate with hands-on experience in full-stack development, component design, REST API integration, and software testing. Skilled in Python with practical experience in Agile development, debugging, and CI/CD. Eager to contribute to reliable software solutions while continuously learning, collaborating with engineering teams, and delivering business value.

Education

- **B.Tech in Computer Science Engineering**, Amrita Vishwa Vidyapeetham, Coimbatore(2022-2026) — graduated
- **Higher Secondary Education**, Ascent Junior College, Vizag(2020-2022) — BIEAP
- **Secondary Education**, Ashram Public School, Kakinada(2020) — CBSE

Work Experience

Full Stack Developer, Indominus Labs Private Limited — Jan 2025 – Jul 2025

- Designed and developed application components using Java, Spring Boot, React.js, REST APIs, MySQL, MongoDB, .NET, and AWS.
- Developed resident, maintenance, billing, and payment modules for a Gated Community Management System.
- Built modular backend services and REST APIs to process, validate, and manage application data.
- Integrated frontend components with backend services and databases to meet functional requirements.
- Analysed application logs, debugged software defects, and improved application reliability.
- Performed functional testing, code reviews, and debugging following Agile development practices.
- Validated API responses, database queries, and application performance to identify and resolve issues.
- Applied Git-based workflows and CI practices for version control, testing, and software integration.
- Collaborated with developers, incorporated feedback, documented features, and delivered assigned components on time.

Projects

- **Swiggy Deployment – Automated Cloud CI/CD Application Pipeline** — GitHub Repo
  Jenkins, Docker, Kubernetes, Terraform, AWS EC2, Linux, Git, SonarQube, Trivy
  - Designed and developed CI/CD components using Jenkins, Docker, Kubernetes, and AWS EC2.
  - Implemented automated build, testing, and deployment workflows following SDLC best practices.
  - Provisioned AWS infrastructure using Terraform and documented technical designs and configurations.
  - Integrated GitHub, Jenkins, SonarQube, Trivy, and Kubernetes into an automated delivery pipeline.
  - Containerized applications using Docker and validated Kubernetes deployments and service integration.
  - Implemented automated code-quality checks and vulnerability scanning using SonarQube and Trivy.
  - Tested and debugged deployment, networking, and application issues using Linux and diagnostic logs.
- **AI StockWise – Financial Data Analytics & AI Platform** — GitHub Repo
  Python, Pandas, Machine Learning, NLP, LLM, RAG, Streamlit, Git, Regression
  - Developed Python components for financial data processing, validation, and analysis using Pandas.
  - Implemented feature engineering and regression models to analyse financial trends and patterns.
  - Developed NLP-based sentiment analysis components using LLMs, RAG, and prompt engineering.
  - Integrated vector embeddings and semantic search for efficient financial information retrieval.
  - Tested and debugged AI components through prompt evaluation, output validation, and model testing.
  - Documented component designs, testing results, model outputs, and implementation workflows.
- **Smart Classroom Automation – Computer Vision Analytics** — GitHub Repo
  Python, OpenCV, YOLOv3-Tiny, Raspberry Pi, ESP32, MQTT, Computer Vision, Git
  - Developed real-time human detection components using Python, OpenCV, and YOLOv3-Tiny.
  - Implemented image processing and occupancy-based automation using Raspberry Pi and ESP32.
  - Performed functional and performance testing to validate detection accuracy and response times.
  - Tested and debugged communication failures, software issues, and system performance.
  - Integrated hardware and software components using MQTT for reliable device communication.
  - Documented component designs, test results, debugging findings, and integration workflows.

Publications

- "Securing Smart Manufacturing IIoT Networks from Advanced Vampire Attack: A Statistical Learning Approach" — ISM 2025, University of Malta. Published in ScienceDirect — Networks, Machine Learning, Cybersecurity
- "A Lightweight Post-Quantum Secure Knapsack Cryptosystem Using Tree Parity Machine Synchronization" — I-SMAC 2025, Published in IEEE-networks,cybersecurity,machine learning

Technical Skills

- **Programming Languages:** Python, Java, JavaScript, SQL, C, C++
- **Software Development:** Component Design, OOP, REST APIs,
- **Frameworks:** Spring Boot, React.js, Node.js, Express.js, Flask
- **Databases:** MySQL, MongoDB
- **Cloud & DevOps:** AWS EC2, Docker, Kubernetes, Jenkins, Terraform, CI/CD
- **Testing & Quality:** Unit Testing, Functional Testing, Integration Testing, Debugging, SonarQube, Trivy
- **System Integration:** REST APIs, MQTT, Data Interfaces, Service Integration
- **AI Tools:** LLMs, RAG, AI Agents, Prompt Engineering, Output Validation
- **Development Tools:** Git, GitHub, Postman, VS Code, Linux
- **Methodologies:** SDLC, Agile Scrum, Technical Documentation, Version Control

Certifications

- Preparing for AWS cloud Certification

Extracurricular Activities

- Central Event Coordinator — CSE Sports Coordinator — Gokulashtami — Taekwondo Black Belt